

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277585

Kind Code

A1

Publication Date

September 04, 2025

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COMBUSTOR SIZE RATING FOR A GAS TURBINE ENGINE USING HYDROGEN FUEL

Abstract

A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length and a burner dome height. The combustion chamber is configured to combust a mixture of the hydrogen fuel flow and the compressed air flow. The combustion chamber can be characterized by a combustor size rating between one inch and seven inches. In more detail, the combustion chamber can be characterized by the combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, in which the combustor size rating is a function of the core air flow parameter.

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Family ID: 96881040

Appl. No.: 18/660525

Filed: May 10, 2024

Related U.S. Application Data

parent US continuation-in-part 17457559 20211203 parent-grant-document US 12078100 child US 18660525

Publication Classification

Int. Cl.: F23R3/28 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation-in-part of U.S. application Ser. No. 17/457,559, filed Dec. 3, 2021, now allowed, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a combustor for a gas turbine engine using hydrogen fuel, and in particular, for a gas turbine engine for aircraft.

BACKGROUND

[0003] The propulsion system for commercial aircraft typically includes one or more aircraft engines, such as turbofan jet engines. The aircraft engine(s) may be mounted to a respective one of the wings of the aircraft, such as in a suspended position beneath the wing using a pylon. These engines may be powered by aviation turbine fuel, which is typically a combustible hydrocarbon liquid fuel, such as a kerosene-type fuel, having a desired carbon number and carbon to hydrogen ratio. Such fuel produces carbon dioxide emissions upon combustion and improvements to reduce such carbon dioxide emissions in commercial aircraft are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Features and advantages of the present disclosure will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

[0005] FIG. 1 is a schematic perspective view of an aircraft having a gas turbine engine according to an embodiment of the present disclosure.

[0006] FIG. 2 is a schematic, cross-sectional view, taken along line 2-2 in FIG. 1, of the gas turbine engine of the aircraft shown in FIG. 1.

[0007] FIG. 3 is a perspective view of an unducted single fan engine that may be used with the aircraft shown in FIG. 1.

[0008] FIG. 4 is a schematic, cross-sectional view, taken along line 4-4 in FIG. 3, of the unducted single fan engine shown in FIG. 3.

[0009] FIG. 5 is a schematic, cross-sectional view, taken along line 2-2 in FIG. 1, of a turbojet engine that may be used with the aircraft shown in FIG. 1.

[0010] FIG. 6 is cross-sectional view of a first combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

[0011] FIG. 7 is cross-sectional view of a second combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

[0012] FIG. 8 is cross-sectional view of a third combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

[0013] FIG. 9 is cross-sectional view of a fourth combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

[0014] FIG. 10 is cross-sectional view of a fifth combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

[0015] FIG. **11** is a graph illustrating combustor length (squared) as a function of combustor height, according to embodiments of the present disclosure.

[0016] FIG. **12** is a graph illustrating combustor size rating as a function of core air flow parameter in engine gas turbine engines using hydrogen fuel, according to embodiments of the present disclosure.

[0017] FIG. **13** depicts a combustion section of a gas turbine engine.

[0018] FIG. **14** is a schematic of a side cross-sectional view taken along line XIV of FIG. **13** of a combustor in the combustion section formed from a combustor liner having multiple sets of dilution passages according to an aspect of the disclosure herein.

[0019] FIG. **15** is a schematic, transverse cross-sectional view of a first dilution passage arrangement provided on a dome wall suitable for use within the combustor of FIG. **14**.

[0020] FIG. **16** is a partial side cross-sectional view of a portion of the first dilution passage arrangement of FIG. **15** as seen from line XVI of FIG. **15**, illustrating a first passage angle defining a first orientation for the dilution passage.

[0021] FIG. **17** is a partial side cross-sectional view of a portion of the first dilution passage arrangement as seen from line XVII of FIG. **15**, illustrating the first passage angle defining a second orientation for the dilution passage.

[0022] FIG. **18** is a partial side cross-sectional view of a portion of the first dilution passage arrangement of FIG. **15** as seen from line XVIII of FIG. **15**, illustrating the first passage angle defining a third orientation for the dilution passage.

[0023] FIG. **19** is an enlarged, schematic, front view of the dome wall as seen from section XIX of FIG. **15**, the dilution passage including a second passage angle.

[0024] FIG. **20** is a schematic, front view of the dome wall including the dilution passage arrangement of FIG. **15**, further illustrating a flame shaping attributable to the dilution passages.

[0025] FIG. **21** is a schematic, transverse view of a second dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a plurality of slots that follow a spiral pattern.

[0026] FIG. **22** is a schematic, transverse view of a third dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a first subset of slots following a curved line and a second subset of slots following a linear line.

[0027] FIG. **23** is a schematic, transverse view of a fourth dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a first subset of slots following a curved line and a second subset of slots following a linear line that is extends non-circumferentially about the dome wall.

[0028] FIG. **24** is a schematic, transverse view of a fifth dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a plurality of slots following a non-circular polygonal path.

[0029] FIG. **25** is a schematic, transverse view of a sixth dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a plurality of slots with a non-symmetrical formation.

[0030] FIG. **26** is a schematic, transverse view of a seventh dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a plurality of slots having differing cross-sectional areas.

[0031] FIG. **27** is a schematic, transverse view of an eighth dilution passage arrangement suitable for use as the dilution passage arrangement of FIG. **15**, the dilution passage arrangement including a plurality of slots having a first row of slots and a second row of slots.

DETAILED DESCRIPTION

[0032] Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, it is to be understood that the following detailed description is exemplary and intended to provide further

explanation without limiting the scope of the disclosure as claimed.

[0033] Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and scope of the present disclosure.

[0034] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0035] The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

[0036] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

[0037] The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.

[0038] The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0039] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a one, two, four, ten, fifteen, or twenty percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values.

[0040] The term “bypass ratio,” unless stated otherwise, means the bypass ratio at take off conditions. The term bypass ratio as used herein means the ratio between the mass flow rate of air flow accelerated by the engine that bypasses the engine core to the mass flow rate of the air flow entering the engine core. For example, in an exemplary engine such as the turbofan engine **100** depicted in FIG. **2** and discussed further below, the bypass ratio is the ratio of the mass flow rate of the air flow entering the bypass air flow passage **140** to the mass flow rate of the air flow entering the core air flow path **121**. The bypass ratio can also be estimated as a ratio of the area of an inlet to the bypass duct (e.g., inlet of the bypass air flow passage **140**, discussed below) or an area swept by a rotor (e.g., the area swept by fan blades **322**, discussed below) to the area of the inlet to the engine core (e.g., inlet of the core air flow path **121**).

[0041] The term “thrust,” unless stated otherwise, means the maximum thrust at take off. This meaning of thrust is adopted when computing a core airflow parameter (relationship (2), below).

[0042] Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0043] Combustible hydrocarbon liquid fuel, such as Jet-A fuel, has long been used in gas turbine engines and the components of gas turbine engines, particularly, the combustor, have been designed for such fuels. A hydrogen fuel may be utilized to eliminate carbon dioxide emissions from commercial aircraft. Hydrogen fuel, however, poses a number of challenges as compared to

combustible hydrocarbon liquid fuel, such as Jet-A fuel. Hydrogen fuel, for example, is a highly reactive fuel that burns at higher temperatures than combustible hydrocarbon liquid fuel. Hydrogen fuel also has much higher flame speeds. For example, the laminar flame speed for a hydrogen fuel of diatomic hydrogen is an order of magnitude greater than the laminar flame speed for Jet-A fuel. [0044] When testing hydrogen fuel in current gas turbine engines with rich burn combustors, we, the inventors, observed that the higher combustion temperature of hydrogen fuel results in increased production of nitrogen oxides (“NO_x”), as compared to combustible hydrocarbon liquid fuel. We also observed in our testing that NO_x emissions are sensitive to combustor residence time. As noted above, hydrogen fuel is highly reactive (relative to other fuels) with a wide range of flammability limits and very high flame speeds, resulting in a very short hydrogen flame close to the front end of the combustor. With such a short flame, the post-flame residence time increases for combustors designed for Jet-A fuel. These findings resulted in a realization that when designing a hydrogen fuel combustor to meet NO_x emission targets, the combustor residence time needs to be reduced by more than about fifty percent. To find a suitable combustor design for gas turbine engines using hydrogen fuel, we conceived of a wide variety of combustors having different shapes and sizes in order to determine which embodiment(s) were most promising for a variety of contemplated engine designs and thrust classes. The various embodiments, as described herein and as shown in the figures, are combustors that are sized to meet NO_x emissions targets.

[0045] FIG. 1 is a perspective view of an aircraft 10 that may implement various preferred embodiments. The aircraft 10 includes a fuselage 12, wings 14 attached to the fuselage 12, and an empennage 16. The aircraft 10 also includes a propulsion system that produces a propulsive thrust required to propel the aircraft 10 in flight, during taxiing operations, and the like. The propulsion system for the aircraft 10 shown in FIG. 1 includes a pair of engines 20. In this embodiment, each engine 20 is attached to one of the wings 14 by a pylon 18 in an under-wing configuration. Although the engines 20 are shown attached to the wing 14 in an under-wing configuration in FIG. 1, in other embodiments, the engine 20 may have alternative configurations and be coupled to other portions of the aircraft 10. For example, the engine 20 may additionally or alternatively include one or more aspects coupled to other parts of the aircraft 10, such as, for example, the empennage 16 and the fuselage 12.

[0046] As will be described further below with reference to FIG. 2, the engines 20 shown in FIG. 1 are gas turbine engines that are each capable of selectively generating a propulsive thrust for the aircraft 10. The amount of propulsive thrust may be controlled at least in part based on a volume of fuel provided to the gas turbine engines 20 via a fuel system 200. The fuel is stored in a fuel tank 212 of the fuel system 200. As shown in FIG. 1, at least a portion of the fuel tank 212 is located in each wing 14 and a portion of the fuel tank 212 is located in the fuselage 12 between the wings 14. The fuel tank 212, however, may be located at other suitable locations in the fuselage 12 or the wing 14. The fuel tank 212 may also be located entirely within the fuselage 12 or the wing 14. The fuel tank 212 may also be separate tanks instead of a single, unitary body, such as, for example, two tanks each located within a corresponding wing 14.

[0047] Although the aircraft 10 shown in FIG. 1 is an airplane, the embodiments described herein may also be applicable to other aircraft 10, including, for example, helicopters and unmanned aerial vehicles (UAV). The aircraft discussed herein are fixed-wing aircraft or rotor aircraft that generate lift by aerodynamic forces acting on, for example, a fixed wing (e.g., wing 14) or a rotary wing (e.g., rotor of a helicopter), and are heavier-than-air aircraft, as opposed to lighter-than-air aircraft (such as a dirigible). The engine 20 may be used in various other applications including stationary power generation systems and other vehicles beyond the aircraft 10 explicitly described herein, such as boats, ships, cars, trucks, and the like.

[0048] FIG. 2 is a schematic, cross-sectional view of one of the engines 20 used in the propulsion system for the aircraft 10 shown in FIG. 1. The cross-sectional view of FIG. 2 is taken along line 2-2 in FIG. 1. For the embodiment depicted in FIG. 2, the engine 20 is a high bypass turbofan engine

that is referred to as a turbofan engine **100** herein. The turbofan engine **100** has an axial direction **A** (extending parallel to a longitudinal centerline axis **101**, shown for reference in FIG. 2), a radial direction **R**, and a circumferential direction. The circumferential direction (not depicted in FIG. 2) extends in a direction rotating about the axial direction **A**. The turbofan engine **100** includes a fan section **102** and a turbomachine **104** disposed downstream from the fan section **102**.

[0049] The turbomachine **104** depicted in FIG. 2 includes a tubular outer housing or nacelle **106** and an inlet **108**. Within the housing **106** there is an engine core, which includes, in a serial flow relationship, a compressor section including a booster or low-pressure (LP) compressor **110** and a high-pressure (HP) compressor **112**, a combustion section **150** (also referred to herein as a combustor **150**), a turbine section including a high-pressure (HP) turbine **116** and a low-pressure (LP) turbine **118**, and a jet exhaust nozzle section **120**. The compressor section, the combustor **150**, and the turbine section together define at least in part a core air flow path **121** extending from the inlet **108** to the jet exhaust nozzle section **120**. The turbofan engine **100** further includes one or more drive shafts. More specifically, the turbofan engine includes a high-pressure (HP) shaft or spool **122** drivingly connecting the HP turbine **116** to the HP compressor **112**, and a low-pressure (LP) shaft or spool **124** drivingly connecting the LP turbine **118** to the LP compressor **110**.

[0050] The fan section **102** shown in FIG. 2 includes a fan **126** having a plurality of fan blades **128** coupled to a disk **130**. The fan blades **128** and the disk **130** are rotatable, together, about the longitudinal centerline axis **101** by the LP shaft **124**. The booster **108** may also be directly driven by the LP shaft **124**, as depicted in FIG. 2. The disk **130** is covered by a rotatable front hub **132** aerodynamically contoured to promote an air flow through the plurality of fan blades **128**. Further, an annular fan casing or outer nacelle **134** is provided, circumferentially surrounding the fan **126** and/or at least a portion of the turbomachine **104**. The nacelle **134** is supported relative to the turbomachine **104** by a plurality of circumferentially spaced outlet guide vanes **136**. A downstream section **138** of the nacelle **134** extends over an outer portion of the turbomachine **104** so as to define a bypass air flow passage **140** therebetween.

[0051] The turbofan engine **100** is operable with the fuel system **200** and receives a flow of fuel from the fuel system **200**. As will be described further below, the fuel system **200** includes a fuel delivery assembly **202** providing the fuel flow from the fuel tank **212** to the turbofan engine **100**, and, more specifically, to a plurality of fuel nozzles **442** that inject fuel into a combustion chamber **430** of the combustor **150**.

[0052] The turbofan engine **100** also includes various accessory systems to aid in the operation of the turbofan engine **100** and/or an aircraft including the turbofan engine **100**. For example, the turbofan engine **100** may include a main lubrication system **171**, a compressor cooling air (CCA) system **173**, an active thermal clearance control (ATCC) system **175**, and a generator lubrication system **177**, each of which is depicted schematically in FIG. 2. The main lubrication system **171** is configured to provide a lubricant to, for example, various bearings and gear meshes in the compressor section, the turbine section, the HP spool **122**, and the LP shaft **124**. The lubricant provided by the main lubrication system **171** may increase the useful life of such components and may remove a certain amount of heat from such components through the use of one or more heat exchangers. The compressor cooling air (CCA) system **173** provides air from one or both of the HP compressor **112** or LP compressor **110** to one or both of the HP turbine **116** or LP turbine **118**. The active thermal clearance control (ATCC) system **175** acts to minimize a clearance between tips of turbine blades and casing walls as casing temperatures vary during a flight mission. The generator lubrication system **177** provides lubrication to an electronic generator (not shown), as well as cooling/heat removal for the electronic generator. The electronic generator may provide electrical power to, for example, a startup electrical motor for the turbofan engine **100** and/or various other electronic components of the turbofan engine **100** and/or an aircraft including the turbofan engine **100**.

[0053] Heat from these accessory systems **171**, **173**, **175**, and **177**, and other accessory systems,

may be provided to various heat sinks as waste heat from the turbofan engine **100** during operation, such as to various vaporizers **220**, as discussed below. Additionally, the turbofan engine **100** may include one or more heat exchangers **179** within, for example, the turbine section or jet exhaust nozzle section **120** for extracting waste heat from an air flow therethrough to also provide heat to various heat sinks, such as the vaporizers **220**, discussed below.

[0054] The fuel system **200** of this embodiment is configured to store the fuel for the turbofan engine **100** in the fuel tank **212** and to deliver the fuel to the turbofan engine **100** via the fuel delivery assembly **202**. The fuel delivery assembly **202** includes tubes, pipes, and the like, to fluidly connect the various components of the fuel system **200** to the turbofan engine **100**. As discussed above, the turbofan engine **100**, and, in particular, the combustor **150** discussed herein may be particularly suited for use with hydrogen fuel (diatomic hydrogen). In the embodiments shown in FIG. 2, the fuel is a hydrogen fuel comprising hydrogen, more specifically, diatomic hydrogen. In some embodiments, the hydrogen fuel may consist essentially of hydrogen.

[0055] The fuel tank **212** may be configured to hold the hydrogen fuel at least partially in the liquid phase and may be configured to provide hydrogen fuel to the fuel delivery assembly **202** substantially completely in the liquid phase, such as completely in the liquid phase. For example, the fuel tank **212** may have a fixed volume and contain a volume of the hydrogen fuel in the liquid phase (liquid hydrogen fuel). As the fuel tank **212** provides hydrogen fuel to the fuel delivery assembly **202** substantially completely in the liquid phase, the volume of the liquid hydrogen fuel in the fuel tank **212** decreases and the remaining volume in the fuel tank **212** is made up by, for example, hydrogen in the gaseous phase (gaseous hydrogen). As used herein, the term “substantially completely” as used to describe a phase of the hydrogen fuel, refers to at least 99% by mass of the described portion of the hydrogen fuel being in the stated phase, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the hydrogen fuel being in the stated phase.

[0056] To store the hydrogen fuel substantially completely in the liquid phase, the hydrogen fuel is stored in the fuel tank **212** at very low (cryogenic) temperatures. For example, the hydrogen fuel may be stored in the fuel tank **212** at about -253 degrees Celsius or less at atmospheric pressure, or at other temperatures and pressures to maintain the hydrogen fuel substantially in the liquid phase. The fuel tank **212** may be made from known materials such as titanium, Inconel®, aluminum, or composite materials. The fuel tank **212** and the fuel system **200** may include a variety of supporting structures and components to facilitate storing the hydrogen fuel in such a manner.

[0057] The liquid hydrogen fuel is supplied from the fuel tank **212** to the fuel delivery assembly **202**. The fuel delivery assembly **202** may include one or more lines, conduits, etc., configured to carry the hydrogen fuel between the fuel tank **212** and the turbofan engine **100**. The fuel delivery assembly **202** thus provides a flow path of the hydrogen fuel from the fuel tank **212** to the turbofan engine **100**. The hydrogen fuel is delivered to the engine by the fuel delivery assembly **202** in the gaseous phase, the supercritical phase, or both (e.g., the gaseous phase and the supercritical phase). The fuel system **200** thus includes a vaporizer **220** in fluid communication with the fuel delivery assembly **202** to heat the liquid hydrogen fuel flowing through the fuel delivery assembly **202**. The vaporizer **220** is positioned in the flow path of the hydrogen fuel between the fuel tank **212** and the turbofan engine **100**. The vaporizer **220** may be positioned at least partially within the fuselage **12** or the wing **14** (both shown in FIG. 1), such as at least partially within the wing **14**. The vaporizer **220** may, however, be positioned at other suitable locations in the flow path of the hydrogen between the fuel tank **212** and the turbofan engine **100**. For example, the vaporizer **220** may be positioned external to the fuselage **12** and the wing **14** (both shown in FIG. 1) and positioned at least partially within the pylon **18** (FIG. 1) or the turbofan engine **100** (FIG. 2). When positioned in the turbofan engine **100**, the vaporizer may be located in the nacelle **134**, for example. Although only one vaporizer **220** is shown in FIG. 2, the fuel system **200** may include multiple vaporizers

220. For example, when a vaporizer **220** is positioned in the turbofan engine **100** or in the pylon **18** and functions as a primary vaporizer configured to operate once the turbofan engine **100** is in a thermally stable condition, another vaporizer **220** is positioned upstream of the primary vaporizer and proximate to the fuel tank **212**, and functions as a primer vaporizer during start-up (or prior to start-up) of the turbofan engine **100**.

[0058] The vaporizer **220** is in thermal communication with at least one heat source **222**, **224**. In this embodiment, the vaporizer **220** is in thermal communication with a primary heat source **222** and an auxiliary heat source **224**. In this embodiment, primary heat source **222** is waste heat from the turbofan engine **100**, and the vaporizer **220** is, thus, thermally connected to at least one of the main lubrication system **171**, the compressor cooling air (CCA) system **173**, the active thermal clearance control (ATCC) system **175**, the generator lubrication system **177**, and the heat exchangers **179** to extract waste heat from the turbofan engine **100** to heat the hydrogen fuel. In such a manner, the vaporizer **220** is configured to operate by drawing heat from the primary heat source **222** once the turbofan engine **100** is capable of providing enough heat, via the auxiliary heat source **224**, to the vaporizer **220**, in order to facilitate operation of the vaporizer **220**.

[0059] The vaporizer **220** may be heated by any suitable heat source, and, in this embodiment, for example, the auxiliary heat source **224** is a heat source external to the turbofan engine **100**. The auxiliary heat source **224** may include, for example, an electrical power source, a catalytic heater or burner, and/or a bleed air flow from an auxiliary power unit. The auxiliary heat source **224** may be integral to the vaporizer **220**, such as when the vaporizer **220** includes one or more electrical resistance heaters, or the like, that are powered by the electrical power source. In this configuration the auxiliary heat source **224** may provide heat for the vaporizer **220** independent of whether or not the turbofan engine **100** is running and can be used, for example, during start-up (or prior to start-up) of the turbofan engine **100**.

[0060] As noted, the vaporizer **220** is in communication with the flow of the hydrogen fuel through the fuel delivery assembly **202**. The vaporizer **220** is configured to draw heat from at least one of the primary heat source **222** and the auxiliary heat source **224** to heat the flow of hydrogen fuel from a substantially completely liquid phase to a substantially completely gaseous phase or to a substantially completely supercritical phase.

[0061] The fuel system **200** also includes a high-pressure pump **232** in fluid communication with the fuel delivery assembly **202** to induce the flow of the hydrogen fuel through the fuel delivery assembly **202** to the turbofan engine **100**. The high-pressure pump **232** may generally be the primary source of pressure rise in the fuel delivery assembly **202** between the fuel tank **212** and the turbofan engine **100**. The high-pressure pump **232** may be configured to increase a pressure in the fuel delivery assembly **202** to a pressure greater than a pressure within the combustion chamber **430** of the combustor **150** of the turbofan engine **100**, and to overcome any pressure drop of the components placed downstream of the high-pressure pump **232**.

[0062] The high-pressure pump **232** is positioned within the flow of hydrogen fuel in the fuel delivery assembly **202** at a location downstream of the vaporizer **220**. In this embodiment, the high-pressure pump **232** is positioned external to the fuselage **12** and the wing **14**, and is positioned at least partially within the pylon **18**, or at least partially within the turbofan engine **100**. More specifically, the high-pressure pump **232** is positioned within the turbofan engine **100**. With the high-pressure pump **232** located in such a position, the high-pressure pump **232** may be any suitable pump configured to receive the flow of hydrogen fuel in substantially completely a gaseous phase or a supercritical phase. In other embodiments, however, the high-pressure pump **232** may be positioned at other suitable locations, including other positions within the flow path of the hydrogen fuel. For example, the high-pressure pump **232** may be located upstream of the vaporizer **220** and may be configured to receive the flow of hydrogen fuel through the fuel delivery assembly **202** in a substantially completely liquid phase.

[0063] The fuel system **200** also includes a metering unit in fluid communication with the fuel

delivery assembly **202**. Any suitable metering unit may be used including, for example, a fuel metering valve **234** placed in fluid communication with the fuel delivery assembly **202**. The fuel delivery assembly **202** is configured to provide the fuel metering valve **234**, and the fuel metering valve **234** is configured to receive hydrogen fuel. In this embodiment, the fuel metering valve **234** is positioned downstream of the high-pressure pump **232**. The fuel metering valve **234** is further configured to provide the flow of the hydrogen fuel to the turbofan engine **100** in a desired manner. The fuel metering valve **234** is configured to provide a desired volume of the fuel at, for example, a desired flow rate, to a fuel manifold **236** of the turbofan engine **100**. The fuel manifold **236** then distributes (provides) the hydrogen fuel received to a plurality of fuel nozzles **442** (see FIG. 6) within the combustion section **150** of the turbofan engine **100** where the hydrogen fuel is mixed with compressed air, and the mixture of hydrogen fuel and compressed air is combusted to generate combustion gases that drive the turbofan engine **100**. Adjusting the fuel metering valve **234** changes the volume of fuel provided to the combustion chamber **430** (see FIG. 6) of the combustor **150** and, thus, changes the amount of propulsive thrust produced by the turbofan engine **100** to propel the aircraft **10**.

[0064] Although the turbofan engine **100** is shown as a direct drive, fixed-pitch turbofan engine **100**, in other embodiments, a gas turbine engine may be a geared gas turbine engine (i.e., including a gearbox between the fan **126** and shaft driving the fan, such as the LP shaft **124**), may be a variable pitch gas turbine engine (i.e., including a fan **126** having a plurality of fan blades **128** rotatable about their respective pitch axes), etc. Additionally, in still other exemplary embodiments, the exemplary turbofan engine **100** may include or be operably connected to any other suitable accessory systems. Additionally, or alternatively, the exemplary turbofan engine **100** may not include or be operably connected to one or more of the accessory systems **171**, **173**, **175**, and **177**, discussed above.

[0065] The turbofan engine **100** discussed herein is an example of the engine **20** in which the combustors **150** discussed herein may be used. In other embodiments, other suitable engines may be utilized with aspects of the present disclosure. For example, FIGS. 3 and 4 show an unducted single fan (USF) engine **300** that may be used as the engine **20** of the aircraft **10** and implement the fuel system described above, and combustor designs discussed further below. FIG. 3 is a perspective view of the USF engine **300** and FIG. 4 is a cross-sectional view taken along a line 4-4 in FIG. 3.

[0066] The USF engine **300** includes a housing **302**. The housing **302** may be formed of a nacelle **310** and spinner **320**. The nacelle **310** and/or the spinner **320** house internal components of the USF engine **300**. For example, the nacelle **310** houses a torque producing system **312** coupled to a shaft **314**. The torque producing system **312** in the embodiments discussed herein is a gas turbine engine, such as the turbomachine **104** discussed above with reference to FIG. 2 and, thus, the nacelle **310** of this embodiment is similar to the tubular outer housing **106** discussed above. As the turbomachine **104** used as the torque producing system **312** of the USF engine has the same or similar components and features as the turbomachine **104** discussed above, a detailed description of the components of the turbomachine **104** used in of the USF engine **300** is omitted.

[0067] The torque producing system **312** and the shaft **314** are configured to operate (e.g., to rotate) the spinner **320**. One or more fan blades **322** are coupled to the spinner **320**. More specifically, the spinner **320** includes a fan hub **324**, and the fan blades **322** are coupled to the fan hub **324**. The spinner **320** rotates with respect to the nacelle **310**. Coupled to the nacelle **310** may be one or more outlet guide vanes **326**. In this embodiment, the outlet guide vanes **326** are positioned aft of the fan blades **322**. During operation, the one or more fan blades **322** (by virtue of the connection to the spinner **320**) rotate circumferentially around a longitudinal centerline **304**, in this embodiment, and the nacelle **310** is stationary such that the one or more outlet guide vanes **326** do not rotate around the longitudinal centerline **304** and are, thus, stationary with respect to rotation about the longitudinal centerline **304**. Although the outlet guide vanes **326** are stationary with respect to the

longitudinal centerline **304**, the outlet guide vanes **326** are capable of being rotated or moved with respect to the nacelle **310**, for example, in the direction A of FIG. 4.

[0068] During operation of the USF engine **300**, air flows from the left side of FIG. 4 toward the right side of FIG. 4. A portion of the air flow may flow past the fan blades **322** and the outlet guide vanes **326**. A portion of the air flow may enter the nacelle **310** through the annular inlet **108** to be mixed with the hydrogen fuel for combustion in a combustor **150** of the USF engine **300** and exit through an outlet **120**. The outlet guide vanes **326** may be movable with respect to the nacelle **310** to guide the air flow in a particular direction. Each outlet guide vane **326** may be movable to adjust the lean, pitch, sweep, or any combination thereof, of the outlet guide vane **326**.

[0069] In the embodiment shown in FIGS. 3 and 4, a forward end or front portion of the housing **302** includes the one or more fan blades **322** and the one or more outlet guide vanes **326**. In other embodiments, the one or more fan blades **322** and the one or more outlet guide vanes **326** may have a different arrangement with respect to the housing **302**. For example, the one or more fan blades **322** and the one or more outlet guide vanes **326** may be located on an aft end or rear portion of the housing **302**, such as coupled to a rear portion of the housing **302**.

[0070] In other embodiments, an engine according to the disclosure may be configured to have either the stationary vanes positioned forward of the rotating blades **322** (thus, the blades **326** are inlet guide vanes) or both the blades **326** and blades **322** configured to operate in a counter-rotating fashion. Either “pusher” or “puller” configurations are contemplated. In each of these alternative embodiments, the fuel delivery system **200** and combustor **150**, as described in great detail below, may be used. An example of a suitable engine configuration for a counter-rotating engine is shown and described in FIG. 1 and col. 3, line 43 through col. 4, line 11 of U.S. Pat. No. 10,800,512, hereby incorporated by reference for all purposes. Alternative embodiments of the USF engine **300** are shown and described in FIGS. 6, 7, and 8 and col. 4, line 51 through col. 5, line 19 of U.S. Pat. No. 10,704,410, hereby incorporated by reference for all purposes.

[0071] In further embodiments, a turbojet engine **350** may be used as the engine **20**. FIG. 5 is a schematic, cross-sectional view of the turbojet engine **350**. The cross-sectional view of FIG. 5 is similar to FIG. 2, which is taken along line 2-2 in FIG. 1. The turbojet engine **350** includes the same or similar components of the turbomachine **104** of the turbofan engine **100** and a detailed description of these components is omitted. An exemplary turbojet engine **350** may not include a fan with bypass duct. An exemplary turbojet engine **350** may have high velocity exhaust from the engine, which produces a majority of the thrust for the turbojet engine **350**. In still further embodiments, other suitable gas turbine engines, such as a turboshaft engine, a turboprop engine, and the like, may be utilized with aspects of the present disclosure.

[0072] As noted above, we conceived of a wide variety of combustors having different shapes and sizes. FIGS. 6 to 10 show various combustor shapes that can suitably be used as the combustor **150** for the gas turbine engines **20** discussed herein. FIGS. 6 to 10 are a detail views showing detail 6 in FIG. 2, and, as FIG. 2 is a cross-sectional view, FIGS. 6 to 10 are also cross-sectional views. FIG. 6 shows a first combustor **401**. FIG. 7 shows a second combustor **403**. FIG. 8 shows a third combustor **405**. FIG. 9 shows a fourth combustor **407**. FIG. 10 shows a fifth combustor **409**.

Although the shapes of these combustors **401**, **403**, **405**, **407**, **409** differ, each of these combustors **401**, **403**, **405**, **407**, **409** has similar components, and common reference numerals are used in FIGS. 6 to 10 to for the same or similar components of these combustors **401**, **403**, **405**, **407**, **409**. Accordingly, the following detailed description of the first combustor **401** also applies to the second combustor **403**, the third combustor **405**, the fourth combustor **407**, and the fifth combustor **409**. Some components, such as the combustor casing **410**, for example, may not be shown in each figure, but such components may nevertheless be applicable to the combustors **403**, **405**, **407**, **409**.

[0073] As shown in FIG. 6, combustor **401** includes a combustor casing **410** and a combustor liner **420**. The combustor casing **410** of this embodiment has an outer casing **412** and an inner casing **414**, and the combustor liner **420** of this embodiment has an outer liner **422** and an inner liner **424**.

A combustion chamber **430** is formed within the combustor liner **420**. More specifically, the outer liner **422** and the inner liner **424** are disposed between the outer casing **412** and the inner casing **414**. The outer liner **422** and the inner liner **424** are spaced radially from each other such that the combustion chamber **430** is defined therebetween. The outer casing **412** and the outer liner **422** form an outer passage **416** therebetween, and the inner casing **414** and the inner liner **424** form an inner passage **418** therebetween. In this embodiment, the combustor **401** is a single annular combustor, but, in other embodiments, the combustor **401** may be any other combustor, including, but not limited to a double annular combustor.

[0074] The combustion chamber **430** has a forward end **432** (downstream end) and an aft end **434** (upstream end). The fuel nozzle **442** is positioned at the forward end **432** of the combustion chamber **430**. The fuel nozzle **442** of this embodiment is part of a swirler/fuel nozzle assembly **440**. In this embodiment, when the combustor **401** is an annular combustor **150**, a plurality of fuel nozzles **442** is arranged in an annular configuration with the plurality of fuel nozzles **442** (the swirler/fuel nozzle assemblies **440**) aligned in a circumferential direction of the combustor **401**.

[0075] As discussed above, the compressor section, the combustor **401**, and the turbine section form, at least in part, the core air flow path **121** extending from the annular inlet **108** to the jet exhaust nozzle section **120**. Air entering through the annular inlet **108** is compressed by blades of a plurality of fans of the LP compressor **110** and HP compressor **112**. A cowl assembly **450** is coupled to the upstream ends of outer liner **422** and the inner liner **424**, respectively. An annular opening **452** formed in the cowl assembly **450** enables compressed air from the compressor section (indicated by arrow B) to enter the combustor **401**. The compressed air flows through the annular opening **452** to support combustion. Another portion of the compressed air flows around the outside of the combustor liner **420** through the outer passage **416** and the inner passage **418**. This air is introduced into the combustion chamber **430** through a plurality of circumferentially spaced dilution holes **426** formed in the combustor liner **420** at positions downstream of the fuel nozzle **442**.

[0076] An annular dome plate **454** extends between, and is coupled to, outer liner **422** and the inner liner **424** near their upstream ends. The plurality of circumferentially spaced swirler/fuel nozzle assemblies **440** is coupled to dome plate **454**. Each swirler/fuel nozzle assembly **440** receives compressed air from the annular opening **452**. The swirler/fuel nozzle assembly **440** includes a swirler **444** that is used to generate turbulence in the air. The fuel nozzle **442** injects fuel into the turbulent air flow and the turbulence promotes rapid mixing of the fuel with the air. The resulting mixture of fuel and compressed air is discharged into combustion chamber **430** and combusted in the combustion chamber **430**, generating combustion gases (combustion products), which accelerate as the combustion gases leave the combustion chamber **430**.

[0077] A turbine nozzle **460** is disposed at the outlet of the combustion chamber **430**. The turbine nozzle **460** may be a stage **1** turbine nozzle. The turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424** at the downstream (aft) ends of each of the outer liner **422** and the inner liner **424**. The turbine nozzle **460** of this embodiment includes an outer band **462** and an inner band **464** coupled to outer liner **422** and the inner liner **424**, respectively. The turbine nozzle **460** also includes a leading edge **466**, which in this embodiment is the location where the turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424**, and the outer band **462** and the inner band **464** each has the leading edge **466**. The turbine nozzle **460** further includes a plurality of circumferentially spaced vanes **468** extending between the outer band **462** and the inner band **464**. The vanes **468** extend in a generally radial direction. The vanes **468**, and the turbine nozzle **460**, is a static component and the vanes **468** may be cured to direct (e.g., spin or swirl) the combustion gases to turn the turbines (e.g., drive the turbine blades) of the first stage of the HP turbine **116**. In this embodiment, the turbine section is a multi-stage turbine and these combustion gases will drive subsequent stages of the HP turbine **116** and the LP turbine **118**. The turbine nozzle **460** may, thus, also be referred to as a stage one nozzle (SiN). As discussed above the HP turbine **116** and the LP

turbine **118**, among other things, drive the LP compressor **110** and HP compressor **112**.

[0078] As noted above, we realized that when designing hydrogen fuel combustor to meet NO_x emission targets, the combustor residence time needs to be reduced. We sized the combustor **401**, and more specifically, the combustor liner **420** for various gas turbine engines and flow rates. These different embodiments are shown below in Table 1 and were developed for different bypass ratios and thrust classes of engines, characterized by the core airflow. In particular, we considered the height H, also referred to as burner dome height, of the combustion chamber **430** and the length L, also referred to as burner length, of the combustion chamber. Diluents could be used to suppress the temperature, and, thus, NO_x production, in the combustion chamber **430** when hydrogen is used as the fuel. With the combustor sized as described in these embodiments, hydrogen fuel can be used without the need of diluents. In some embodiments, no diluent is added to the combustion chamber **430** and the fuel is substantially completely diatomic hydrogen without diluent. As used herein, the term “substantially completely,” as used to describe the amount of a particular element or molecule (e.g., diatomic hydrogen), refers to at least 99% by mass of the described portion of the element or molecule, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the element or molecule.

[0079] FIGS. **6** to **10** illustrate how the height H and length L may be determined for the different shapes of combustion liners **420** shown in these figures. The height H of the combustion chamber **430** is taken at the forward end **432** of the combustion chamber **430**. The height H is the maximum height between an inner surface of the outer liner **422** and an inner surface of the inner liner **424** at the forward end **432** of the combustion chamber **430**. The height H is measured along a line (referred to as a forward line **472**, herein) that is generally orthogonal the inner surfaces of the outer liner **422** and the inner liner **424**. The forward line **472** may be orthogonal to a central axis **477** of the fuel nozzle assembly **440** and/or the fuel nozzle **442**. In this manner, the height H may be orthogonal to the central axis **477**. In some embodiments, the height H measured using with the forward line is the maximum height of the combustion chamber **430** and may also be the maximum dome height of the combustion chamber **430**.

[0080] The length L of the combustion chamber **430** is the distance between forward line **472** and the leading edge **466** of the turbine nozzle **460**. As with the height H, a line (referred to as the aft line **474**, herein) can be drawn from the leading edge **466** at the outer liner **422** and leading edge at the inner liner **424**. Each of the forward line **472** and the aft line **474** has a midpoint (midpoint **476** and midpoint **478**, respectively) that is halfway between the outer liner **422** and the inner liner **424**. The length L can be measured from the midpoint **476** of the forward line **472** to the midpoint **478** of the aft line **474**. The midpoint **478** may be the midspan height of the turbine nozzle **460**.

[0081] When developing a gas turbine engine, the interplay between components can make it particularly difficult to select or to develop one component during engine design and prototype testing, especially, when some components are at different stages of completion. For example, one or more components may be nearly complete, yet one or more other components may be in an initial or preliminary phase such that only one (or a few) design parameters are known. It is desired to arrive at what is possible at an early stage of design, so that the down selection of candidate optimal designs, given the tradeoffs, become more possible. Heretofore, the process has sometimes been more ad hoc, selecting one design or another without knowing the impact when a concept is first taken into consideration. For example, various aspects of the fan **126** design, the HP compressor **112** design, and/or the LP compressor **110** design may not be known, but such components impact the core air flow through the core air flow path **121**, and, thus, may influence the design of the combustion chamber **430**.

[0082] We desire to narrow the range of configurations or combination of features that can yield favorable results given the constraints of the design, feasibility, manufacturing, certification requirements, etc., early in the design selection process to avoid wasted time and effort. During the

course of the evaluation of different embodiments as set forth above, we, the inventors, discovered, unexpectedly, that there exists a relationship between the burner length and the burner dome height, which uniquely identifies a finite and readily ascertainable (in view of this disclosure) number of embodiments suitable for a particular architecture that can meet NOx emissions for hydrogen fuel and provide desired flame residence times. This relationship is referred to by the inventors as the combustor size rating (CSR) (in), and is defined according to the following relationship (1) between burner length L (in) and burner dome height H (in):

$$[00001] \text{ CombustorSizeRating(CSR)} = (L)^2 / (H) \quad (1)$$

[0083] As discussed further below, we have identified a range of the Combustor Size Ratings that enable a combustion chamber **430** to be designed for a gas turbine engine **20** using hydrogen fuel. This relationship is applicable over a wide range of thrust class and engine designs. Using this unique relationship, a combustor **150** design can be developed early in the design process that meets NOx emissions targets and reduces engine weight for gas turbine engines using hydrogen fuel.

[0084] Table 1 describes exemplary embodiments 1 to 24 identifying the CSR for various hydrogen fuel burning engines. The embodiments 1 to 24 may be engines with either rich burn combustors or lean burn combustors. Each of embodiments 1 to 24 burns hydrogen fuel. Embodiments 1 to 24 may represent any of the engines described with respect to FIGS. **1** to **5** and can be applied to any of the combustion chamber **430** shapes shown in FIGS. **6** to **10**. In Table 1, the CSR is determined based on the relationship (1) described above. A core air flow parameter (CAFP) (kN) is defined according to the following relationship (2) between thrust (kN) and bypass ratio, both at take off.

$$[00002] \text{ CoreAirFlowParameter} = \frac{\text{Thrust}}{\text{BypassRatio}} \quad (2)$$

The burner length is the length L identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two inches and six inches. In embodiments 1 to 24, the burner length squared may be between six square inches and thirty-five square inches. The burner dome height is the height H identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two and one half inches and six inches.

TABLE-US-00001 TABLE 1 Combustor Core Air Flow Size Rating Parameter Thrust Bypass																								
Embodiment	(in)	(kN)	(kN)	Ratio	1	4.30	38.16	332.39	8.71	2	6.67	49.85	254.26	5.10	3	6.67	53.44	272.53	5.10	4	6.67	51.18	261.03	
5	6.67	52.36	267.03	5.10	6	4.69	21.07	120.10	5.70	7	4.69	23.80	121.40	5.10	8	3.01	12.58	64.53	5.13	9	3.01	12.18	62.49	
5.13	10	3.00	16.44	83.70	5.09	11	3.00	15.20	82.10	5.40	12	3.00	14.27	84.20	5.90	13	3.00	14.27	84.20	5.90	14	3.00	14.27	
84.20	5.90	15	2.12	36.55	321.60	8.80	16	2.12	39.23	345.20	8.80	17	2.12	40.65	349.20	8.59	18	2.12	40.65	349.20	8.59	19	2.12	
37.34	299.81	8.03	20	1.67	13.63	143.10	10.50	21	1.94	51.51	489.30	9.50	22	5.51	40.89	363.90	8.90	23	2.46	12.72	147.28	11.58	24	2.70
5.00	150.00	30.00																						

[0085] The length L may be between 2.63 inches and 5.60 inches. The length L may be between two inches and three inches. The length L may be between two and one half inches and three and one half inches. The height H may be between 2.80 inches and 5.60 inches. The height H may be between two and one half inches and six inches. The height H may be between two and one half inches and five inches. The height H may be between four inches and five inches. The burner length squared may be between 6.89 inches and 31.36 inches. The burner length squared may be between six square inches and thirty-five square inches. The burner length squared may be between six square inches and twenty square inches. The burner length squared may be between six square inches and twelve square inches. The burner length squared may be between eight square inches and twelve square inches. The burner length squared and the height may be any values such that the CSR is less than seven inches. The burner length squared and the height may be any values such that the CSR is less than six inches.

[0086] FIG. **11** represents, in graph form, the burner length, squared, as a function of the burner dome height. FIG. **11** shows that the burner length, squared, may be changed based on the burner

dome height. An area **500** may present the boundaries of burner length, squared, as a function of burner dome height in which a particular combustor is designed. FIG. **12** represents, in graph form, the CSR as a function of core air flow parameter. Table 1 and FIG. **12** show that CSR may be changed based on a thrust class, as characterized by the core air flow parameter, of an engine. An area **600** may present the boundaries of CSR as a function of the core air flow parameter in which a particular combustor is designed.

[0087] As shown in FIG. **12**, the CSR is less than seven inches for every core air flow. That is, the CSR is less than seven inches for every thrust class of engine. The CSR may be between 1.67 inches and 6.67 inches. The CSR may be between one inch and seven inches. The CSR may be between one and one half inches and seven inches. The CSR may be between two inches and seven inches. The CSR may be between two inches and six inches. The CSR may be between one inches and five inches. The CSR may be between two inches and five inches. The CSR may be between three inches and five inches. The core air flow parameter may be less than sixty kN. The core air flow parameter may be between five kN and 53.44 kN. The core air flow parameter may be between two and one half kN and sixty kN. The core air flow parameter may be between ten kN and twenty kN. The core air flow parameter may be between thirty kN and forty-five kN.

[0088] With continued reference to FIG. **12**, the CSR may be a function of the core air flow parameter. The CSR may be based on a thrust of the gas turbine engine. The CSR may be between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN. The CSR may be between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN. The thrust may be between sixty kN and five hundred kN. The thrust may be between 62.49 kN and 489.30 kN. The CSR is defined by a relationship of the burner length, squared, and the burner dome height.

[0089] In an extension of the concepts disclosed hereinabove, also provided herein is a combustor with the dome wall having fuel cups with a dilution passage arrangement. A dilution passage arrangement is provided around each fuel cup and the dilution passage arrangement of each fuel cup can be selected to function with adjacent fuel cups and their corresponding dilution passage arrangement to collectively control the annular flame spread from all of the fuel cups as well as individually controlling the flame spread from each fuel cup. Each dilution passage arrangement includes a set of dilution passages terminating in a plurality of slots provided along the dome wall. As described herein, a single "dilution passage arrangement" refers to a plurality of slots provided around a single, corresponding fuel cup of the set of fuel cups.

[0090] It was determined that hydrogen fuel results in higher combustion temperatures as compared to combustible hydrocarbon liquid fuel and that NO_x emissions are sensitive to factors including combustor residence times. The present disclosure allows for alternate flame shaping and dilution to allow for combustors that are sized to meet NO_x emissions targets.

[0091] The various embodiments of the combustors, as described herein and shown in the figures, include the plurality of slots that create a layer of insulation (e.g., the curtain of compressed air) between the ignited hydrogen fuel and the dome wall, the inner liner, the outer liner, and any portions of the gas turbine engine outside of the dome wall, the inner liner and the outer liner. In these high burn temperatures it is desirable to have additional insulation between the ignited hydrogen fuel and surrounding components of the gas turbine engine.

[0092] The curtain of compressed air is further used to shape the flame within the combustion chamber, which in turn results in an enhanced control of the flame shape profile. By shaping the flame the liner wall temperature, the dome wall temperature, the combustor exit temperature profile and pattern of the flame/gas exiting the combustor is controlled. This control or shaping can further ensure that the combustion section or otherwise hot sections of the turbine engine do not fail or otherwise become ineffective by being overly heated, thus increasing the lifespan of the turbine engine. Further, the introduction of the dilution passage arrangements, as described herein, ensure an even, uniform, or otherwise desired flame propagation within the combustor.

[0093] Referring now to FIG. 13 a view of a portion of a combustion section 714 of an engine can be seen. It will be understood that the combustion section 714 is similar to the combustion sections previously described and can include any features described above even if not identified further herein. A set of fuel cups 776 are disposed around a combustor centerline 736. The combustor centerline 729 can be the centerline 721 of the engine. The combustor centerline 736 can be a centerline for the combustion section 714, a single combustor, or a set of combustors that are arranged about the combustor centerline 736.

[0094] The combustor 780 can have a can, can-annular, or annular arrangement depending on the type of engine in which the combustor 780 is located. In a non-limiting example, an annular arrangement is illustrated and disposed within a casing 738. The combustor 780 is defined by a combustor liner 7872 including an outer annular combustor liner 782a and an inner annular combustor liner 782b concentric with respect to each other and annular about the combustor centerline 736. A dome assembly 784 including a dome wall 790 together with the combustor liner 782 can define a combustion chamber 786 annular about the combustor centerline 736. At least one fuel cup 736, illustrated as multiple fuel injectors annularly arranged about the combustor centerline 736, is fluidly coupled to the combustion chamber 786. A compressed air passageway 788 can be defined at least in part by both the combustor liner 782 and the casing 738.

[0095] The at least one fuel cup 736 is included within a plurality of fuel cups 736. Each fuel cup 736 can include a fuel cup centerline 734 that extends into the page. Each fuel cup centerline 734 can be arranged along a circumferential line 730. Alternatively, one or more fuel cups 736 can be offset from the circumferential line 730. Additionally, the fuel cups 736 can be arranged such that the fuel cup centerlines 734 form a pattern relative to, but not necessarily on, the circumferential line 730.

[0096] Each fuel cup centerline 734 in combination with the combustor centerline 736, can be used to define a respective fuel cup reference line 730 that extends radially from the combustor centerline 736 and through the corresponding fuel cup centerline 734. For the purposes of illustration, four fuel cup reference lines 730 are shown, however, it will be appreciated that each fuel cup 736 includes a fuel cup reference line 730. The fuel cup reference line 730 is used in this description to establish a local polar coordinate system 732 for each fuel cup 736. The local polar coordinate system defines a 0-180 degree line lying on the corresponding reference line 730, and a 90-270 degree line for each of the four illustrated fuel cup reference lines 730. The 0 degree and 90 degree lines have been shown for convenience on each of the polar coordinate systems 732. Since the fuel cups 736 are circumferentially spaced around the combustor centerline 736, a polar coordinate system based on the fuel cup reference line 730 is a convenient way to describe the local fuel cups 736, while taking into account the rotational shifts in the local coordinate system due to the circumferential arrangement.

[0097] FIG. 14 illustrates first set of dilution passages 792, a second set of dilution passages 793 and a third set of dilution passages 794 can fluidly connect the compressed air passageway 788 and the combustor 780.

[0098] The fuel cup 736 can be coupled to and disposed within the dome assembly 784. The fuel cup 736 can include a flare cone 804 and a swirler 812. The flare cone 804 includes an outlet 796 of the fuel cup 736 directly fluidly coupled to the combustion chamber 786. The fuel cup 736 is fluidly coupled to a fuel inlet 798 via a linear passageway 800.

[0099] Both the inner and outer combustor liners 782a, 782b can have an outer surface 806 and an inner surface 808 at least partially defining the combustion chamber 786. The combustor liner 782 can be made of one continuous monolithic portion or be multiple monolithic portions assembled together to define the inner and outer combustor liners 782a, 782b. By way of non-limiting example, the outer surface 806 can define a first piece of the combustor liner 782 while the inner surface 808 can define a second piece of the combustor liner 782 that when assembled together form the combustor liner 782. As described herein, the combustor liner 782 includes the third set of

dilution passages **794**. It is further contemplated that the combustor liner **782** can be any type of combustor liner **782**, including but not limited to a single wall or a double walled liner or a tile liner. An ignitor **810** can be provided at the combustor liner **782** and fluidly coupled to the combustion chamber **786**, at any location, by way of non-limiting example upstream of the third set of dilution passages **794**.

[0100] During operation, a compressed air (C) can flow from the compressor section **714** to the combustor **780** through the dome assembly **784**. The compressed air (C) is fed to the fuel cup **736** via the swirler **812** as a swirled airflow (S). A flow of fuel (F) is fed to the fuel cup **736** via the fuel inlet **798** and the linear passageway **800**. The swirled airflow (S) and the flow of fuel (F) are mixed at the flare cone **804** and fed to the combustion chamber **786** as a fuel/air mixture. The ignitor **810** can ignite the fuel/air mixture to define a flame within the combustion chamber **786**, which generates a combustion gas (G). While shown as starting axially downstream of the outlet **796**, it will be appreciated that the fuel/air mixture can be ignited at or near the outlet **796**.

[0101] The compressed air (C) is further fed to dilution passages **792**, **793** as a first dilution airflow (D1) and to the third set of dilution passages **794** as a second dilution airflow (D2). The first dilution airflow (D1) is used to direct and shape the flame, while the second dilution airflow (D2) is used to direct the combustion gas (G).

[0102] The combustor **780** shown in FIG. **14** is well suited for the use of a hydrogen-containing gas as the fuel because it helps contain the faster moving flame front associated with hydrogen fuel, as compared to traditional hydrocarbon fuels.

[0103] FIG. **15** is a schematic, transverse, cross-sectional view of a first dilution passage arrangement **900** on a dome wall **902** suitable for use within any of the previously described combustors. Therefore, similar parts of the first dilution passage arrangement **900** will be given similar names, with it being understood that the description applies to the first dilution passage arrangement **900**, unless indicated otherwise. The first dilution passage arrangement **900** is provided on the dome wall **902** around a fuel cup **904** having a fuel cup centerline **910** and an outlet **905**. The dome wall **902** extends between an outer liner **906** and an inner liner **908**.

[0104] A plurality of dilution passages **912** extend through the dome wall **902** and include a plurality of slots **914**. Each slot of the plurality of slots **914** defines a termination point of one or more dilution passages **912** of the plurality of dilution passages **912**. Each dilution passage **912** extends along a passage centerline **934** that terminates at a respective slot **914** to define a center point (indicated by the passage centerline **934** on each dilution on each slot **914**) of the respective slot **914**. The plurality of slots **914** are circumferentially spaced about at least a portion of the fuel cup centerline **910**. As a non-limiting example, a single dilution passage **912** terminates in a single slot **914**. However, a dilution passage can have multiple branches, with each branch terminating in a slot. Each slot of the plurality of slots **914** is defined by a cross-sectional area when viewed along a vertical plane extending perpendicularly to the fuel cup centerline **910** and intersecting the slot **914**. The cross-sectional area can be any suitable shape such as, but not limited to, obround, ovate, oblong, round, elongated, rectangular, triangular, or the like. Further, the cross-sectional area can be uniform or non-uniform amongst the plurality of slots **914** such that one or more of the slots can be larger or include a different shape than another slot.

[0105] At least a portion of the plurality of slots **914** are arranged such that the passage centerline **934** is provided along a first line **916**. Another portion of the plurality of slots **914** are arranged such that their passage centerlines **934** are provided along a second line **918**. As illustrated, the first line **916** and the second line **918** are arcs centered on the fuel cup centerline **910**. Some of these additional paths are illustrated in the different arrangements shown in FIGS. **16-27**.

[0106] The first dilution passage arrangement **900** can be positioned about the fuel cup **904** with respect to a polar coordinate system **969**. The polar coordinate system **969** includes a 0 degree to 180 degree line defining a fuel cup reference line **972**, and a 90 degree to 270 degree line defining a transverse reference line **970**. The polar coordinate system **969** can be divided into four quadrants:

a first quadrant **974** between 0-90 degrees, a second quadrant **976** between 90-180 degrees, a third quadrant **978** between 180-270 degrees and a fourth quadrant **980** between 270 to 360 degrees. [0107] The first line **916** and the second line **918** each define arc segments extending circumferentially around at least a portion of the fuel cup centerline **910**. These arc segments are defined as slot-present arc segments. A first break **920** and a second break **922** are formed circumferentially between the first line **916** and the second line **918**. The first break **920** and the second break **922** define opposing slot-free arc segments. The first break **920** is provided within ± 75 degrees of the transverse reference line **970**. The second break **922** is provided within ± 75 degrees of the transverse reference line **970**.

[0108] The first line **916** and the second line **918**, and thus the plurality of slots **914**, can extend across or within any suitable portion of the polar coordinate system **969**. As a non-limiting example, the first line **916** or the second line **918**, and thus the plurality of slots **914**, can extend between at least two adjacent quadrants.

[0109] The first dilution passage arrangement **900** is symmetrical or non-symmetrical about at least one of the transverse reference line **970** or the fuel cup reference line **972**.

[0110] During operation, a fuel/air mixture (F1) is supplied through the outlet **905** of the fuel cup **904**. The fuel/air mixture (F1) can exit the fuel cup **904** in a straight line or otherwise include a circumferential swirl, thus defining the fuel/air mixture (F1) as a swirled fuel/air mixture. When swirled, the fuel/air mixture (F1) includes a circumferential component, with respect to the fuel cup centerline **910**. The plurality of slots **914** circumscribe at least a portion of the fuel air mixture (F1).

[0111] FIGS. **16-18** illustrate various non-limiting configurations of the plurality of dilution passages **912** extending through the dome wall **902**. Each dilution passage **912** extends between an inlet **928** and a respective slot **914**. The passage centerline **934** extends linearly or non-linearly. The fuel cup **904** includes a flare cone **930** with a flared surface **932** opening up to the outlet **905**. The dome wall **902**, the outer liner **906** and the inner liner **908** (FIG. **15**) at least partially define a combustion chamber **924**. The outlet **905** of the fuel cup **904** and the slot **914** of the dilution passage **912** are each directly fluidly coupled to respective portions of the combustion chamber **924**. It will be appreciated that the dilution passage **912** can take any suitable form and include any other suitable structure. As a non-limiting example, the inlet **928** can flare outwardly to define a funnel or otherwise include a chute that extends axially from the dome wall **902**, with respect to the passage centerline **934**.

[0112] FIG. **16** illustrates a partial cross-sectional side view of a dilution passage **912** of the plurality of dilution passages **912**. The passage centerline **934** of the illustrated dilution passage **912** extends parallel to the fuel cup centerline **910** forming an axial dilution passage.

[0113] The passage centerline **934**, specifically where the passage centerline **934** at the slot **914** (e.g., the center point of the slot **914**), is provided a first radial height (Rh1) from the fuel cup centerline **910**. The slot **914** is defined by a slot width (Sw). The dilution passage **912** extends from the inlet **928** to the slot **914** a total axial length (La), with respect to the fuel cup centerline **910**. The outlet **905** of the fuel cup **904** extends a second radial height (Rh2) from the fuel cup centerline **910**. The outlet **905**, as a non-limiting example, is circular such that the second radial height (Rh2) is a radius of the outlet **905** and that two times the second radial height (Rh2) is the width of the outlet **905**.

[0114] A ratio between the second radial height (Rh2) and the first radial height (Rh1) is greater than or equal to 1 and less than or equal to 3. A ratio of the slot width (Sw) to the width of the outlet **905** (e.g., two times the second radial height (Rh2)) is greater than or equal to 0.03 and less than or equal to 0.5. The slot width (Sw) can be any suitable size such as greater than or equal to 0.04 inches. A ratio between the total axial length (La) to the slot width (Sw) can be greater than or equal to 0.1 and less than or equal to 10.

[0115] It has been found that conforming the first dilution passage arrangement **900** and the fuel cup **904** to the above-described ratios and ranges provides a distinct benefit when compared to a

dilution passage arrangement **900** and fuel cup **904** that does not fall within the aforementioned ratios and ranges. These benefits will be described later in the specification with respect to FIG. **20**. [0116] FIG. **17** illustrates a partial cross-sectional side view of a dilution passage **912** of the plurality of dilution passages **912**. The passage centerline **934** of the illustrated dilution passage **912** extends radially outward from the fuel cup centerline **910** forming an outward dilution passage. The passage centerline **934** forms a first passage angle (β) with respect to a projection **936** of the fuel cup centerline **910**.

[0117] FIG. **18** illustrates a partial cross-sectional side view of a dilution passage **912** of the plurality of dilution passages **912**. The passage centerline **934** of the illustrated dilution passage **912** extends radially inward towards the fuel cup centerline **910** forming an inward dilution passage. The passage centerline **934** forms a first passage angle (β) with respect to the projection **936** of the fuel cup centerline **910**.

[0118] The first passage angle (β) can be any suitable angle that is greater than or equal to negative 70 degrees and less than or equal to 70 degrees.

[0119] While illustrated as the plurality of dilution passages **912** including the axial dilution passages **912**, the outward dilution passages **912** and the inward dilution passages **912**, it will be appreciated that the plurality of dilution passages **912** can be formed as only axial dilution passages **912**, only outward dilution passages **912**, only inward dilution passages **912**, or any suitable combination thereof.

[0120] FIG. **19** is an enlarged schematic front view of the dome wall **902** as seen from section XIX of FIG. **15**. As illustrated, the dilution passage **912** includes a respective passage centerline **934** that forms a second passage angle (θ) with respect to a projection **971** of the transverse reference line **970** (FIG. **15**). The second passage angle (θ) can have an absolute value of greater than or equal to 0 degrees and less than or equal to 90 degrees. As a non-limiting example, the absolute value of the second passage angle (θ) of at least a portion of the dilution passages **912** can be greater than or equal to 0 degrees and less than or equal to 30 degrees. It will be further appreciated that at least a portion of the dilution passages **912** can be formed without a second passage angle (θ) such that they extend into the illustrated page and coincide with, or otherwise circumscribe, the slot **914**.

[0121] A slot airflow (F_s) can flow outward from the slot **914**. The slot airflow (F_s) can include the second passage angle (θ) at the slot **914**. As such, the slot airflow (F_s) can be defined by a circumferential component, with respect to the fuel cup centerline **910**. The circumferential component of the slot airflow (F_s) can be in line with/parallel with, or counter to/non-parallel with the circumferential component of the fuel air mixture (F_1) (FIG. **15**).

[0122] FIG. **20** is a schematic front view of the dome wall **902** of FIG. **15** having the same view of FIG. **15**. The dilution passage arrangement **900** includes a slot-present region **913** extending between opposing breaks **920**, **922**. Any number of one or more slots of the plurality of slots **914** (FIG. **15**) are provided within each slot-present region **913**. During operation, the fuel air mixture (F_1) is ignited to define a flame **940**, and a compressed airflow is fed through the plurality of dilution passages **912**. The compressed airflow forms a curtain around at least a portion of the circumferential extent of the flame **940**. The flame **940**, however, is free to flow through the first break **920** and the second break **922** in the directions indicated by arrows **942**, **944**, respectively.

[0123] A plurality of fuel cups **904** (FIG. **15**) are circumferentially arranged about the dome wall **902**. Each fuel cup **904** can include a respective first dilution passage arrangement **900**. The dilution passage arrangements **900** can be the same or different between fuel cups **904**. It is contemplated that the first break **920** of a first dilution passage arrangement **900** can be at least partially aligned with a second break **922** of a second dilution passage arrangement **900** that is circumferentially adjacent to the first dilution passage arrangement **900**. The flame **940** that spreads through the first break **920** of the first dilution passage arrangement **900** can meet with and merge with the flame **940** that spreads through the second break **922** of the second dilution passage arrangement **900**. This merging flames **940** ensures that a continuous annular ring of flame is

formed along the dome wall **902**, which ensures flame propagation from one fuel cup **904** to another and reduces the likelihood of a flameout at any given one of the fuel cups **904**.

[0124] The flow of compressed air flowing through the slots **914** (FIG. **15**) can be defined by a total slot flow. The fuel air mixture (F1) can further be defined by a total fuel cup flow. The total slot flow and the total fuel cup flow are each defined by a volume of fluid (e.g., compressed air or fuel/air mixture, respectively) that flows through the respective slots **914** or fuel cup **904** (FIG. **15**) over a period of time (e.g., milliliters/second). The ratio between the total slot flow and the total fuel cup flow can be greater than or equal to 0.2 and less than or equal to 4.

[0125] The curtain of compressed air from the dilution passages **912** is used for a multitude of reasons. First, the curtain of compressed air prevents the flame **940** from contacting or otherwise overly heating the dome wall **902**, the outer liner **906** and the inner liner **908**. This, in turn, ensures that that dome wall **902**, the outer liner **906**, the inner liner **908** or any portions of the combustor or gas turbine engine outside of the dome wall **902**, the inner liner **908** or the outer liner **906** are not damaged or otherwise overly heated by the flame **940**. Second, the curtain of compressed air is used to shape the flame **940**. The flame shaping can be done, in part, by the first passage angle (β) (FIGS. **17** and **18**) or the second passage angle (θ) (FIG. **19**). For example, an outward dilution passage **912** (FIG. **17**) will allow the flame **940** to expand, thereby generating a flame **940** with a larger surface area, while an inward dilution passage **912** (FIG. **18**) will compress or constrict the flame **940**, thereby generating a flame **940** with a smaller surface area.

[0126] Further, the orientation of or the inclusion of the second passage angle (θ) can be used to provide a hydrodynamic curtain of compressed air oriented with respect to the fuel air mixture (F1). It has been found that the orientation of the curtain of compressed air can be used to shape and direct the flame **940**. As a non-limiting example, when the circumferential component of the curtain of compressed air is non-parallel to the circumferential component of the fuel/air mixture (F1), the curtain of compressed air is better adapted to directing the flame **940** away from the outer liner **906** and the inner liner **908**. As a non-limiting example, when the circumferential component of the curtain of compressed air is parallel to the circumferential component of the fuel air mixture (F1), the curtain of compressed air is better adapted to directing the flame **940** away from the dome wall **902**. When the fuel/air mixture (F1) does not include a circumferential component, the curtain of compressed air is used to swirl the fuel/air mixture in a desired fashion.

[0127] The curtain of compressed air can further be used to ensure that the combustor including the first dilution passage arrangement **900** can use fuels with high burn temperatures, and burning at fast flame speeds, such as hydrogen-containing fuels. As hydrogen-containing fuels have a significantly higher burn temperature than traditional hydrocarbon fuels, it becomes more important to insulate the flame **940** from the dome wall **902**, the outer liner **906** and the inner liner **908** and to cool the dome wall **902**, the outer liner **906** and the inner liner **908**. The air curtain that is generated through the first dilution passage arrangement **900** is used to provide a layer of insulation (e.g., the curtain of compressed air) between the flame **940** and the dome wall **902**, the outer liner **906** and the inner liner **908** and to cool the dome wall **902**, the outer liner **906** and the inner liner **908** and to direct the flame **940** away from the dome wall **902**, the outer liner **906** and the inner liner **908**.

[0128] It is contemplated that the ratio of Rh2:Rh1 being greater than or equal to 1 and less than or equal to 3 results in a desirable shaping of the flame **940** that does not overly heat the outer liner **906**, the inner liner **908** or the dome wall **902** while still having desired properties of the flame **940**. For example, if the ratio of Rh2:Rh1 were greater than 3, it has been found that the flame **940** expands radially outward, with respect to the fuel cup centerline **910** (FIG. **15**), and impinges the inner liner **908** and outer liner **906**, resulting in an overheating of the inner liner **908** and outer liner **906**. If, however, the ratio of Rh2:Rh1 were less than 1, it has been found that the flame **940** is too centered around a central region defined by the fuel cup centerline **910**, which in turn results in an undesirable profile and pattern of the flame **940**.

[0129] It is contemplated that the ratio of the slot width (Sw) to the width of the outlet **905** being greater than or equal to 0.03 and less than or equal to 0.3 results in a plurality of slots **914** that have a sufficient flow rate of compressed air with respect to a flow rate of the fuel and air mixture (F1) flowing from the fuel cup **904** in order to produce a desirable shape of the flame **940**. If the ratio of the of the slot width (Sw) to the width of the outlet **905** were larger than 0.3, it has been found that too much compressed air exits the plurality of slots **914**, resulting in the flame **940** having too high of a velocity or otherwise being overly compressed. If, however, the ratio of the slot width (Sw) to the width of the outlet **905** is smaller than 0.03, it has been found that the compressed air exiting the plurality of slots **914** is not sufficient in creating in the curtain of compressed air that insulates the dome wall **902**, the outer liner **906** and the inner liner **908** from the heat of the flame **940**, nor does the curtain of compressed air have enough force to shape the flame **940** in the desired pattern.

[0130] It is contemplated that the ratio between the total axial length (La) to the slot width (Sw) being greater than or equal to 0.1 and less than or equal to 10 results in a desired velocity of the compressed air exiting the plurality of slots **914**. For example, if the ratio between the total axial length (La) to the slot width (Sw) were greater than 10, the total axial length (La) is longer, meaning that the compressed air flowing through the dilution passage **912** will frictional losses, which ultimately lowers the kinetic energy, as opposed to a lower total axial length (La). This reduction in the kinetic energy due to frictional losses ultimately results in a combustor with unsatisfactory performance when compared to a combustor falling within the desired total axial length (La) to slot width (Sw) ratio. If, however, the ratio between the total axial length (La) to the slot width (Sw) were less than 0.1, it has been found that the losses (e.g., windage losses) associated with the compressed air entering the combustion chamber and merging with the fuel and air mixture (F1) within the combustion chamber. These losses ultimately results in a combustor with unsatisfactory performance when compared to a combustor falling within the desired total axial length (La) to slot width (Sw) ratio.

[0131] FIG. **21** is a schematic, transverse cross-sectional view of an exemplary second dilution passage arrangement **1500** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The second dilution passage arrangement **1500** is similar to the first dilution passage arrangement **900**, therefore, like parts will be identified by like numerals increased to the **1500** series, with it being understood that the description of the first dilution passage arrangement **900** applies to the second dilution passage arrangement **1500**, unless otherwise noted.

[0132] The second dilution passage arrangement **1500** is provided on a dome wall **1502** and surrounding a fuel cup **1504** having a fuel cup centerline **1510**. The dome wall **1502** extends radially between an outer liner **1506** and an inner liner **1508**. A plurality of dilution passages **1512** extend through the dome wall **1502** and terminate in a plurality of slots **1514** formed along the dome wall **1502**. The second dilution passage arrangement **1500** is provided along a polar coordinate system **1569** having a fuel cup reference line **1572** extending from 0 degrees to 180 degrees and a transverse reference line **1570** extending from 90 degrees to 270 degrees. The plurality of slots **1514** extend along, at least, a first line **1516** and a second line **1518**.

[0133] The second dilution passage arrangement **1500** is similar to the first dilution passage arrangement **900**, except that the first line **1516** and the second line **1518** each serially increase in a radial distance from the fuel cup centerline **1510** from a first slot **1515** of the plurality of slots **1514** to a second slot **1517** of the plurality of slots **1514**. The first slot **1515** and the second slot **1517** can define circumferential ends of the first line **1516** or the second line **1518**, such that the radial distance increases from one circumferential end (e.g., the first slot **1515** or the second slot **1517**) to a second circumferential end (e.g., another of the first slot **1515** or the second slot **1517**) along the respective first line **1516** or the second line **1518**. The first slot **1515** and the second slot **1517** being provided on circumferentially opposite ends of the first line **1516** and the second line **1518**. The first slot **1515** is a first radial distance **1558** from the fuel cup centerline **1510**. The second slot **1517** is a second radial distance **1560** from the fuel cup centerline **1510**. The first radial distance

1558 is smaller than the second radial distance **1560**. The radial distances of the slots **1514** circumferentially between the first slot **1515** and the second slot **1517** can increase serially in a continuous or non-continuous fashion such that the first radial distance **1558** is the smallest and the second radial distance **1560** is the largest. This configuration can, for example, form a spiral pattern of slots **1514** along the dome wall **1502**.

[0134] The benefit of including the spiral pattern, or serially-increasing radial heights, is that the second dilution passage arrangement **1500** can be used to further shape the flame (e.g., the flame **940** of FIG. **21**) that exits the fuel cup **1504** similar to how the second passage angle (e.g., the second passage angle (θ) of FIG. **19**) is used to shape the flame. For example, the first line **1516** and the second line **1518** can form a spiral that extends circumferentially (e.g., from the first slot **1515** to the second slot **1517**) parallel to or counter to the circumferential direction of the fuel air mixture (e.g., the fuel air mixture (**F1**)) leaving the fuel cup **1504**.

[0135] FIG. **22** is a schematic, transverse cross-sectional view of an exemplary third dilution passage arrangement **1600** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The third dilution passage arrangement **400** is similar to the dilution passage arrangement **200**, **1500** (FIG. **21**), therefore, like parts will be identified by like numerals increased to the **1600** series, with it being understood that the description of the dilution passage arrangement **900**, **1500** applies to the third dilution passage arrangement **1600**, unless otherwise noted.

[0136] The third dilution passage arrangement **1600** is provided on a dome wall **1602** and surrounding a fuel cup **1604** having a fuel cup centerline **1610**. The dome wall **1602** extends radially between an outer liner **1606** and an inner liner **1608**. A plurality of dilution passages **1612** extend through the dome wall **1602** and terminate in a plurality of slots **1614** formed along the dome wall **1602**. The third dilution passage arrangement **1600** is provided along a polar coordinate system **1669** having a fuel cup reference line **1672** extending from 0 degrees to 180 degrees and a transverse reference line **1670** extending from 90 degrees to 270 degrees. The plurality of slots **1614** extend along, at least, a first line **1616** and a second line **1618**. The third dilution passage arrangement **1600** includes a first break **1620** and a second break **1622**.

[0137] The third dilution passage arrangement **1600** is similar to the dilution passage arrangement **900**, **1500**, except that the first line **1616** and the second line **1618** each include at least two non-parallel or non-uniform portions. As a non-limiting example, the first line **1616** and the second line **1618** each include a curved line **1662** and at least one linear line **1664**. The curved line **1662** can include two circumferential ends and the at least one linear line **1664** can extend outwardly from one of the two circumferential ends of the curved line **1662**. As a non-limiting example, the first line **1616** and the second line **1618** each include the curved line **1662** and the at least one linear line **1664** including a first linear line extending from a first circumferential end of the curved line **1662** and a second linear line extending from a second circumferential end of the curved line **1662**. The two linear lines **1664** can each be formed identical or non-identical to one another. The linear lines **1664** can extend parallel with or non-parallel to the transverse reference line **1670**. There can be any number of one or more slots on the curved line **1662** and the linear lines **1664** of the first line **1616** and the second line **1618**.

[0138] The linear lines **1664** can correspond to an extend along the first break **1620** and the second break **1622** and extend radially outward from, with respect to the fuel cup centerline **1610**, respective portions of the corresponding curved line **1662**. As such, the linear lines **1664** can form a channel for the first break **1620** and the second break **1622**.

[0139] The slots **1614** provided on the first line **1616** and the second line **1618** can each be defined by a respective cross-sectional area. The cross-sectional area of the slots **1614** can be equal to or non-equal to the cross-sectional area of the slots **1614** on the second line **1618**. The cross-sectional area of the slots **1614** can be constant or differ along the respective first line **1616** and the second line **1618**. As a non-limiting example, the slots **1614** on the curved line **1662** can have a circular cross-sectional area while the slots **1614** on the linear lines **1664** can have an oblong cross-

sectional area. The oblong cross-sectional area can be used to create elongated troughs extending along the first break **1620** and the second break **1622**.

[0140] FIG. **23** is a schematic, transverse view of an exemplary fourth dilution passage arrangement **1700** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The fourth dilution passage arrangement **1700** is similar to the dilution passage arrangement **900**, **1500** (FIG. **21**), **1600** (FIG. **22**), therefore, like parts will be identified by like numerals increased to the **1700** series, with it being understood that the description of the dilution passage arrangement **900**, **1500**, **1600** applies to the fourth dilution passage arrangement **1700**, unless otherwise noted.

[0141] The fourth dilution passage arrangement **1700** is provided on a dome wall **1702** and surrounding a fuel cup **1704** having a fuel cup centerline **1710**. The dome wall **1702** extends radially between an outer liner **1706** and an inner liner **1708**. A plurality of dilution passages **1712** extend through the dome wall **1702** and terminate in a plurality of slots **1714** formed along the dome wall **1702**. The fourth dilution passage arrangement **1700** is provided along a polar coordinate system **1769** having a fuel cup reference line **1772** extending from 0 degrees to 180 degrees and a transverse reference line **1770** extending from 90 degrees to 270 degrees. The plurality of slots **1714** extend along, at least, a first line **1716** and a second line **1718**. The fourth dilution passage arrangement **1700** includes a first break **1720** and a second break **1722**.

[0142] The fourth dilution passage arrangement **1700** is similar to the third dilution passage arrangement **1600** (FIG. **22**) in that it includes the first line **1716** and the second line **1718**, each defined by a curved line **1762** and at least one linear line **1764** (e.g., two linear lines **1764**). The difference, however, is that the at least one linear line **1764** extends at an angle **1766** with respect to the transverse reference line **1770**. An absolute value of the angle **1766** can be greater than or equal to 0 degrees and less than or equal to 60 degrees.

[0143] With reference to FIGS. **22** and **23**, the benefit of the dilution passage arrangements **1600**, **1700** is further directing and shaping of the flame (e.g., the flame **940** of FIG. **21**). For example, the channel formed by the linear lines **1664**, **1764** can be used to direct or otherwise channel the flame from one dilution passage arrangement **1600**, **1700** to another, circumferentially adjacent dilution passage arrangement **1600**, **1700**.

[0144] FIG. **24** is a schematic, transverse view of an exemplary fifth dilution passage arrangement **1800** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The fifth dilution passage arrangement **1800** is similar to the dilution passage arrangement **900**, **1500** (FIG. **21**), **1600** (FIG. **22**), **1700** (FIG. **23**), therefore, like parts will be identified by like numerals increased to the **1800** series, with it being understood that the description of the dilution passage arrangement **900**, **1500**, **1600**, **1700** applies to the fifth dilution passage arrangement **1800**, unless otherwise noted.

[0145] The fifth dilution passage arrangement **1800** is provided on a dome wall **1802** and surrounding a fuel cup **1804** having a fuel cup centerline **1810**. The dome wall **1802** extends radially between an outer liner **1806** and an inner liner **1808**. A plurality of dilution passages **1812** extend through the dome wall **1802** and terminate in a plurality of slots **1814** formed along the dome wall **1802**. The fifth dilution passage arrangement **1800** is provided along a polar coordinate system **1869** having a fuel cup reference line **1872** extending from 0 degrees to 180 degrees and a transverse reference line **1870** extending from 90 degrees to 270 degrees. The plurality of slots **1814** extend along, at least, a first line **1816** and a second line **1818**.

[0146] For purposes of illustration, the first line **1816** and the second line **1818** are projected outwardly beyond outside slots **1819** of the plurality of slots **1814** along the first line **1816** and the second line **1818** based on a trajectory of the first line **1816** and second line **1818** at the outside slots **1819**. As illustrated, the first line **1816** and the second line **1818** meet to form a continuous polygonal path about the fuel cup centerline **1810**. The polygonal path can be any suitable circular or non-circular path.

[0147] The first line **1816** and the second line **1818** can meet at two location along the transverse reference line **1870**. Alternatively, the first line **1816** and the second line **1818** can meet at any

suitable location along the dome wall **1802**.

[0148] FIG. **25** is a schematic, transverse view of an exemplary sixth dilution passage arrangement **1900** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The sixth dilution passage arrangement **1900** is similar to the dilution passage arrangement **900**, **1500** (FIG. **21**), **1600** (FIG. **22**), **1700** (FIG. **23**), **1800** (FIG. **24**), therefore, like parts will be identified by like numerals increased to the **1900** series, with it being understood that the description of the dilution passage arrangement **900**, **1500**, **1600**, **1700**, **1800** applies to the sixth dilution passage arrangement **1900**, unless otherwise noted.

[0149] The sixth dilution passage arrangement **1900** is provided on a dome wall **1902** and surrounding a fuel cup **1904** having a fuel cup centerline **1910**. The dome wall **1902** extends radially between an outer liner **1906** and an inner liner **1908**. A plurality of dilution passages **1912** extend through the dome wall **1902** and terminate in a plurality of slots **1914** formed along the dome wall **1902**. The sixth dilution passage arrangement **1900** is provided along a polar coordinate system **1969** having a fuel cup reference line **1972** extending from 0 degrees to 180 degrees and a transverse reference line **1970** extending from 90 degrees to 270 degrees. The plurality of slots **1914** extend along, at least, a first line **1916** and a second line **1918**.

[0150] The sixth dilution passage arrangement **1900** is similar to the dilution passage arrangement **900**, **1500**, **1600**, **1700**, **1800**, except that the sixth dilution passage arrangement **1900** is non-symmetrical about both the transverse reference line **1970** and the fuel cup reference line **1972**. The sixth dilution passage arrangement **1900** can further include a plurality of slots **1914** that are non-uniformly and non-equally distributed along the dome wall **1902**. In other words, the sixth dilution passage arrangement **1900** can include any random or non-random distribution of slots **1914** along the dome wall **1902**. As a non-limiting example, at least one slot **1914** can lay along a tangent line **1977** extending from the fuel cup **904**.

[0151] FIG. **26** is a schematic, transverse view of an exemplary seventh dilution passage arrangement **2000** suitable for use as the first dilution passage arrangement **900** of FIG. **15**. The seventh dilution passage arrangement **2000** is similar to the dilution passage arrangement **900**, **1500** (FIG. **10**), **1600** (FIG. **11**), **1700** (FIG. **12**), **1800** (FIG. **13**), **1900** (FIG. **14**), therefore, like parts will be identified by like numerals increased to the 2000 series, with it being understood that the description of the dilution passage arrangement **900**, **1500**, **1600**, **1700**, **1800**, **1900** applies to the seventh dilution passage arrangement **2000**, unless otherwise noted.

[0152] The seventh dilution passage arrangement **2000** is provided on a dome wall **2002** and surrounding a fuel cup **2004** having a fuel cup centerline **2010**. The dome wall **2002** extends radially between an outer liner **2006** and an inner liner **2008**. A plurality of dilution passages **2012** extend through the dome wall **2002**. The seventh dilution passage arrangement **2000** is provided along a polar coordinate system **2069** having a fuel cup reference line **2072** extending from 0 degrees to 180 degrees and a transverse reference line **2070** extending from 90 degrees to 270 degrees.

[0153] The set of dilution passages **2012** terminate in a first group of slots **2082** and a second group of slots **2084**, each disposed on a first line **2016** and a second line **2018**. The first group of slots **2082** can have a different formation with respect to the second group of slots **2084**. As a non-limiting example, each slot of the first group of slots **2082** can include a cross-sectional area that is larger than or smaller than a cross-sectional area of each slot of the second group of slots **2084**. As a non-limiting example, each slot of the second group of slots **2084** can include a second passage angle (e.g., the second passage angle (θ) of FIG. **20**) while each slot of the first group of slots **2082** do not.

[0154] The first group of slots **2082** and the second group of slots **2084** can each be continuously provided on a suitable portion of the first line **2016** and the second line **2018**. As a non-limiting example, there can be two separate groups of the second group of slots **2084** per the first line **2016** and second line **2018**. As a non-limiting example, the second group of slots **2084** can be provided

along circumferentially distal ends of the first line **2016** and second line **2018**. It will be appreciated that the seventh dilution passage arrangement **2000** can include any number of two or more groups of slots.

[0155] The benefit of including the seventh dilution passage arrangement **2000** having the first group of slots **2082** and the second group of slots **2084** is that the seventh dilution passage arrangement **2000** allows for tuning of the flame shape and cooling/insulation efficiency of the seventh dilution passage arrangement **2000**. As a non-limiting example, the second group of slots **2084** can be provided along circumferentially distal ends of the first line **2016** and the second line **2018** and include the second passage angle. The first group of slots **2082** can be provided circumferentially between the second group of slots **2084** and be inwardly, outwardly, or axial slots without a second passage angle. As such, the second group of slots **2084** can be used to provide the hydrodynamic curtain of air that is in-line with or counter to the fuel air mixture, as described herein, while the first group of slots **2082** can be used to compress or expand the flame.

[0156] FIG. **27** is a schematic, transverse view of an exemplary eighth dilution passage arrangement **2100** suitable for use as the first dilution passage arrangement **2100** of FIG. **15**. The eighth dilution passage arrangement **2100** is similar to the dilution passage arrangement **900**, **1500** (FIG. **10**), **1600** (FIG. **11**), **1700** (FIG. **12**), **1800** (FIG. **13**), **1900** (FIG. **14**), **2000** (FIG. **15**), therefore, like parts will be identified by like numerals increased to the **2100** series, with it being understood that the description of the dilution passage arrangement **900**, **1500**, **1600**, **1700**, **1800**, **1900**, **2000** applies to the eighth dilution passage arrangement **2100**, unless otherwise noted.

[0157] The eighth dilution passage arrangement **2100** is provided on a dome wall **2102** and surrounding a fuel cup **2104** having a fuel cup centerline **2110**. The dome wall **2102** extends radially between an outer liner **2106** and an inner liner **2108**. A plurality of dilution passages **2112** extend through the dome wall **2102**. The eighth dilution passage arrangement **2100** is provided along a polar coordinate system **2169** having a fuel cup reference line **2172** extending from 0 degrees to 180 degrees and a transverse reference line **2170** extending from 90 degrees to 270 degrees. The polar coordinate system **2169** includes a first quadrant **2174**, a second quadrant **2176**, a third quadrant **2178**, and a fourth quadrant **2180**.

[0158] The eighth dilution passage arrangement **2100**, like the seventh dilution passage arrangement **2000**, can include the first group of slots **2182** and the second group of slots **2184** disposed on a first line **2116** and a second line **2118**. The difference, however, is that the eighth dilution passage arrangement **2100** further includes a third group of slots **2186** provided radially outward, with respect to the fuel cup centerline **2110**, from the first group of slots **2182** and the second group of slots **2184**. This, in turn, forms a first row of slots **2190** having the first group of slots **2182** and the second group of slots **2184** following the first line **2116** and the second line **2118**, and a second row of slots **2192** having the third group of slots **2186** and following a third line **2188**.

[0159] The third group of slots **2186** can be uniform or non-uniform with the first group of slots **2182** or the second group of slots **2184**. The third line **2188** can extend linearly or non-linearly and be parallel to or non-parallel to the transverse reference line **2170**. The second row of slots **2192** can include a total of four third groups of slots **2186**. As a non-limiting example, one third group of slots **2186** can be provided in each of the first quadrant **2174**, the second quadrant **2176**, the third quadrant **2178** and the fourth quadrant **2180**.

[0160] The benefit of having the first row of slots **2190** and the second row of slots **2192** is to further shape the flame (e.g., the flame **940** of FIG. **15**) such that the flame does not escape through the curtain of compressed air generated by the first row of slots **2190** and the second row of slots **2192** and heat the outer liner (e.g., the outer liner **906** of FIG. **15**), inner liner (e.g., the inner liner **908** of FIG. **25**) or dome wall **2102**.

[0161] Benefits of the present disclosure include a combustor suitable for use with a hydrogen-containing fuel. As outlined previously, hydrogen-containing fuels have a higher flame temperature

than traditional fuels (e.g., fuels not containing hydrogen). That is, hydrogen or a hydrogen mixed fuel typically has a wider flammable range and a faster burning velocity than traditional fuels such as petroleum-based fuels, or petroleum and synthetic fuel blends. These high burn temperatures of hydrogen-containing fuel mean that additional insulation is needed between the ignited hydrogen-containing fuel and surrounding components of the gas turbine engine (e.g., the dome wall, the inner/outer liner, and other parts of the gas turbine engine). The combustor, as described herein, includes the plurality of slots that create a layer of insulation (e.g., the curtain of compressed air) between the ignited hydrogen-containing fuel and the dome wall, the inner liner, the outer liner, and any portions of the gas turbine engine outside of the dome wall, the inner liner and the outer liner. The curtain of compressed air is further used to shape the flame within the combustion chamber, which in turn results in an enhanced control of the flame shape profile. By shaping the flame the liner wall temperature, the dome wall temperature, the combustor exit temperature profile and pattern of the flame/gas exiting the combustor can be controlled. This control or shaping can further ensure that the combustion section or otherwise hot sections of the turbine engine do not fail or otherwise become ineffective by being overly heated, thus increasing the lifespan of the turbine engine. Further, the introduction of the dilution passage arrangements, as described herein, ensure an even, uniform, or otherwise desired flame propagation within the combustor.

[0162] Benefits associated with using hydrogen-containing fuel over conventional fuels include an eco-friendlier engine as the hydrogen-containing fuel, when combusted, generates less carbon pollutants than a combustor using conventional fuels. For example, a combustor including 100% hydrogen-containing fuel (e.g., the fuel is 100% H₂) would have zero carbon pollutants. The combustor, as described herein, can be used in instances where 100% hydrogen-containing fuel is used.

[0163] Further benefits associated with using hydrogen-containing fuel over conventional fuels include a gas turbine engine that can utilize less fuel due to higher heating value of fuel to achieve same turbine inlet temperatures. For example, a conventional gas turbine engine using conventional fuels will require more fuel to produce the same amount of work or engine output as the present gas turbine engine using hydrogen-containing fuels. This, in turn, means that either less amount of fuel can be used to generate the same amount of engine output as a conventional gas turbine engine, or the same amount of fuel can be used to generate an excess of increased engine output when compared to the conventional gas turbine engine.

[0164] To the extent not already described, the different features and structures of the various embodiments can be used in combination, or in substitution with each other as desired. That one feature is not illustrated in all of the embodiments is not meant to be construed that it cannot be so illustrated, but is done for brevity of description. Thus, the various features of the different embodiments can be mixed and matched as desired to form new embodiments, whether or not the new embodiments are expressly described. All combinations or permutations of features described herein are covered by this disclosure.

[0165] Further aspects of the present disclosure are provided by the subject matter of the following clauses.

[0166] A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length L and a burner dome height H, the combustion chamber configured to combust a mixture of the hydrogen fuel flow and the compressed air flow, and the combustion chamber being characterized by a combustor size rating between one inch and seven inches.

[0167] The gas turbine engine of the preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein H is the maximum height between the inner surface of the outer liner

and the inner surface of the inner liner at the forward end of the combustion chamber.

[0168] The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.

[0169] The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.

[0170] The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.

[0171] The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.

[0172] The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.

[0173] The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.

[0174] The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.

[0175] The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.

[0176] The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.

[0177] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.

[0178] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.

[0179] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.

[0180] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.

[0181] The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.

[0182] The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.

[0183] The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein L is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

[0184] The gas turbine engine of any preceding clause, further comprising one or more rotating blades.

[0185] The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.

[0186] The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.

[0187] The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.

[0188] The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.

[0189] The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.

[0190] The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.

[0191] The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.

[0192] The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.

[0193] The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.

[0194] The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.

[0195] An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.

[0196] The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.

[0197] A gas turbine engine including a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter.

[0198] The gas turbine engine of any preceding clause, wherein the core air flow parameter is a relationship between the thrust and bypass ratio.

[0199] The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

[0200] The gas turbine engine of any preceding clause, wherein the combustor size rating is based on a thrust of the gas turbine engine.

[0201] The gas turbine engine of any preceding clause, wherein the thrust is between sixty kN and five hundred kN.

[0202] The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.

[0203] The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein the burner length is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

[0204] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.

[0205] The gas turbine engine of any preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein the burner dome height is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.

[0206] The gas turbine engine of any preceding clause, wherein the combustor size rating is

between two inches and seven inches.

[0207] The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.

[0208] The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.

[0209] The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.

[0210] The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.

[0211] The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.

[0212] The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.

[0213] The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.

[0214] The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.

[0215] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.

[0216] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.

[0217] The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.

[0218] The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.

[0219] The gas turbine engine of any preceding clause, further comprising one or more rotating blades.

[0220] The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.

[0221] The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.

[0222] The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.

[0223] The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.

[0224] The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.

[0225] The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.

[0226] The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.

[0227] The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.

[0228] The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the

turbojet engine not including a fan with bypass duct.

[0229] The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.

[0230] An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.

[0231] The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.

[0232] A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor defining a combustor centerline and including a set of fuel cups circumferentially spaced relative to the combustor centerline, with each fuel cup having a fuel cup centerline, a set of dilution passages for each fuel cup of the set of fuel cups, with each dilution passage of the set of dilution passages having a passage centerline, and a plurality of slots spaced about a fuel cup in the set of fuel cups, with each slot of the plurality of slots defining a termination of at least one dilution passage of the set of dilution passages and including a center point defined as a location where the passage centerline of the at least one dilution passage intersects the slot, a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height and wherein the core air flow parameter is a relationship between the thrust and bypass ratio.

[0233] The gas turbine engine of any preceding clause, wherein the center points of the plurality of slots are located on a polar coordinate system having a fuel cup reference line extending through the fuel cup centerline and defining a 0 degree to 180 degree reference line, with 0 degrees being radially closest to the combustor centerline, a transverse reference line defining a 90 degree to 270 degree reference line, a first quadrant extending between 0 degrees and 90 degrees, a second quadrant extending between 90 degrees and 180 degrees, a third quadrant extending between 180 degrees and 270 degrees, and a fourth quadrant extending between 270 degrees and 360 degrees, wherein opposing breaks are defined by slot-free arc segments between ± 75 degrees from the transverse reference line, and opposing slot-present arc segments are located between the slot-free arc segments, with the plurality of slots being located in the slot-present arc segments and not present in the slot-free arc segments.

[0234] The gas turbine engine of any preceding clause, wherein opposing breaks are defined by slot-free arc segments between ± 75 degrees from the transverse reference, and opposing slot-present arc segments are located between the slot-free arc segments, with the plurality of slots being located in the slot-present arc segments and not present in the slot-free arc segments.

[0235] The gas turbine engine of any preceding clause, wherein the passage centerline forms a first passage angle with respect to the fuel cup centerline, with the first passage angle being greater than or equal to -70 degrees and less than or equal to 70 degrees.

[0236] The gas turbine engine of any preceding clause, wherein the set of dilution passages includes a first dilution passage having a first passage angle and a second dilution passage having a first passage angle, non-equal to the first passage angle of the first dilution passage.

[0237] The gas turbine engine of any preceding clause, wherein each dilution passage includes a respective first passage angle that is non-equal to each of the other first passage angles.

[0238] The gas turbine engine of any preceding clause, wherein the plurality of slots includes a first

row of slots provided along a first line and a second row of slots positioned radially outward from the first row of slots, and being provided along a second line, separate from the first line.

[0239] The gas turbine engine of any preceding clause, wherein the second line is linear.

[0240] The gas turbine engine of any preceding clause, wherein the second line forms a 0 degree angle with respect to the transverse reference line.

[0241] The gas turbine engine of any preceding clause, wherein the second line forms an angle with respect to the transverse reference line having an absolute value greater than 0 degrees and less than or equal to 60 degrees.

[0242] The gas turbine engine of any preceding clause, wherein the first line is non-linear.

[0243] The gas turbine engine of any preceding clause, wherein the first passage angle of the plurality of slots provided on the first line are larger than the first passage angle of the plurality of slots provided on the second line.

[0244] The gas turbine engine of any preceding clause, wherein the plurality of slots extend between at least two adjacent quadrants.

[0245] The gas turbine engine of any preceding clause, wherein the plurality of slots are provided along a first line, and each slot of the plurality of slots on the first line is provided a radial distance from the fuel cup centerline, with the radial distance serially increasing from one circumferential end of the first line to another circumferential end of the first line.

[0246] The gas turbine engine of any preceding clause, wherein a first subset of the plurality of slots follow a curved line, and a second subset of the slots follow a linear line extending from a circumferential end of the curved line, with the linear line corresponding to at least one of the opposing breaks.

[0247] The gas turbine engine of any preceding clause, wherein the linear line forms an angle with respect to the transverse reference line, the angle having an absolute value of greater than or equal to 0 degrees and less than or equal to 70 degrees.

[0248] The gas turbine engine of any preceding clause, wherein the plurality of slots are non-symmetrical about the corresponding fuel cup reference line.

[0249] The gas turbine engine of any preceding clause, wherein the plurality of slots follow a first line and a second line, separate from the first line, with the first line and the second line being symmetrical or non-symmetrical about at least one of either the fuel cup reference line or the transverse reference line.

[0250] The gas turbine engine of any preceding clause, wherein the set of dilution passages form a dilution passage arrangement about each fuel cup of the set of fuel cups, with at least two dilution passage arrangements being different from each other.

[0251] The gas turbine engine of any preceding clause, wherein the plurality of slots follow a non-circular polygonal path that extends circumferentially about the fuel cup centerline.

[0252] The gas turbine engine of any preceding clause, wherein the set of fuel cups receive a flow of fuel including a hydrogen-containing fuel.

[0253] The gas turbine engine of any preceding clause, wherein the center point of each slot of the plurality of slots is located a first radial height from the fuel cup centerline, each fuel cup of the set of fuel cups includes an outlet formed along the dome wall, with a radially outer surface of the outlet located a second radial height from the fuel cup centerline, and a ratio between the first radial height and the second radial height is greater than 1 and less than or equal to 3.

[0254] The gas turbine engine of any preceding clause, wherein each slot of the plurality of slots includes a slot width when viewed along a vertical plane perpendicular to the corresponding fuel cup centerline and intersecting a respective slot, each fuel cup of the set of fuel cups includes an outlet formed along the dome wall, the outlet having an outlet width, and a ratio between the slot width and the outlet width is greater than or equal to 0.03 and less than or equal to 0.5.

[0255] The gas turbine engine of any preceding clause, wherein a fuel/air mixture is fed to the combustion chamber, through the set of fuel cups, with a portion of the fuel/air mixture being fed

through a corresponding fuel cup of the set of fuel cups at a fuel/air volumetric flow rate, and a compressed air is fed to the combustion chamber through the corresponding set of dilution passages at a compressed air volumetric flow rate, with a ratio between the fuel/air volumetric flow rate and the compressed air volumetric flow rate being greater than or equal to 0.2 and less than or equal to 4.

[0256] The gas turbine engine of any preceding clause, wherein each dilution passage of the plurality of dilution passages includes a total axial length between an inlet of the dilution passage and a respective slot, the respective slot includes a slot width when viewed along a vertical plane perpendicular to the corresponding fuel cup centerline and intersecting the respective slot, and with a ratio between the total axial length and the slot width being greater than or equal to 0.1 and less than or equal to 10.

[0257] Although the foregoing description is directed to the preferred embodiments, it is noted that other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or scope of the disclosure. Moreover, features described in connection with one embodiment may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A gas turbine engine comprising: a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow; a compressor section configured to compress air flowing therethrough to provide a compressed air flow; and a combustor configured to operate without diluent, the combustor defining a combustor centerline and comprising: an inner liner; an outer liner; a set of fuel cups circumferentially spaced relative to the combustor centerline, with each fuel cup having a fuel cup centerline; a set of dilution passages for each fuel cup of the set of fuel cups, with each dilution passage of the set of dilution passages having a passage centerline; a plurality of slots spaced about a fuel cup in the set of fuel cups, with each slot of the plurality of slots defining a termination of at least one dilution passage of the set of dilution passages and including a center point defined as a location where the passage centerline of the at least one dilution passage intersects the slot; and a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by:

$L \cdot \sup. 2/H$ wherein H is a maximum height of the combustion chamber measured by a forward line extending from an inner surface of the outer liner to an inner surface of the inner liner and L is a length of the combustion chamber measured from a midpoint of the forward line to a midpoint of an aft line, the aft line extending from the inner surface of the inner liner to the inner surface of the outer liner at a leading edge of a turbine nozzle, and, wherein the core air flow parameter is defined by:

Thrust/Bypass Ratio.

2. The gas turbine engine of claim 1, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

3. The gas turbine engine of claim 1, wherein the combustor size rating is based on a thrust of the gas turbine engine.

4. The gas turbine engine of claim 3, wherein the thrust is between sixty kN and five hundred kN.

5. The gas turbine engine of claim 1, further comprising a turbine nozzle downstream of the combustion chamber, wherein the burner length is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

6. The gas turbine engine of claim 5, wherein the burner length, squared, is between six square inches and thirty-five square inches.

7. The gas turbine engine of claim 1, wherein the center points of the plurality of slots are located on a polar coordinate system having: a fuel cup reference line extending through the fuel cup centerline and defining a 0 degree to 180 degree reference line, with 0 degrees being radially closest to the combustor centerline; a transverse reference line defining a 90 degree to 270 degree reference line; a first quadrant extending between 0 degrees and 90 degrees; a second quadrant extending between 90 degrees and 180 degrees; a third quadrant extending between 180 degrees and 270 degrees; and a fourth quadrant extending between 270 degrees and 360 degrees; wherein opposing breaks are defined by slot-free arc segments between ± 75 degrees from the transverse reference line, and opposing slot-present arc segments are located between the slot-free arc segments, with the plurality of slots being located in the slot-present arc segments and not present in the slot-free arc segments.
8. The gas turbine engine of claim 7, wherein the passage centerline forms a first passage angle with respect to the fuel cup centerline, with the first passage angle being greater than or equal to -70 degrees and less than or equal to 70 degrees.
9. The gas turbine engine of claim 1, wherein the set of dilution passages includes a first dilution passage having a first passage angle and a second dilution passage having a first passage angle, non-equal to the first passage angle of the first dilution passage.
10. The gas turbine engine of claim 1, wherein each dilution passage includes a respective first passage angle that is non-equal to each of the other first passage angles.
11. The gas turbine engine of claim 1, wherein the plurality of slots includes a first row of slots provided along a first line and a second row of slots positioned radially outward from the first row of slots, and being provided along a second line, separate from the first line.
12. The gas turbine engine of claim 11, wherein a first passage angle of the plurality of slots provided on the first line are larger than a first passage angle of the plurality of slots provided on the second line.
13. The gas turbine engine of claim 1, wherein the plurality of slots are provided along a first line, and each slot of the plurality of slots on the first line is provided a radial distance from the fuel cup centerline, with the radial distance serially increasing from one circumferential end of the first line to another circumferential end of the first line.
14. The gas turbine engine of claim 1, wherein a first subset of the plurality of slots follow a curved line, and a second subset of the slots follow a linear line extending from a circumferential end of the curved line, with the linear line corresponding to at least one of the opposing breaks.
15. The gas turbine engine of claim 1, wherein the plurality of slots are non-symmetrical about a corresponding fuel cup reference line.
16. The gas turbine engine of claim 1, wherein the set of fuel cups are provided on a burner dome.
17. The gas turbine engine of claim 1, wherein the plurality of slots follow a first line and a second line, separate from the first line, with the first line and the second line being symmetrical or non-symmetrical about at least one of either a fuel cup reference line or a transverse reference line.
18. The gas turbine engine of claim 1, wherein the set of dilution passages form a dilution passage arrangement about each fuel cup of the set of fuel cups, with at least two dilution passage arrangements being different from each other.
19. The gas turbine engine of claim 1, wherein the plurality of slots follow a non-circular polygonal path that extends circumferentially about the fuel cup centerline.
20. The gas turbine engine of claim 1, wherein the set of fuel cups receive a flow of fuel including a hydrogen-containing fuel.
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